

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

Let the median and the mean deviation about the median of 7 observation 170, 125, 230, 190, 210, a, b be 170 and $\frac{205}{7}$ respectively. Then the mean deviation about the mean of these 7 observations is :

- (1) 31
- (2) 28
- (3) 30
- (4) 32

Q2 - 2024 (01 Feb Shift 2)

Consider 10 observation $x_1, x_2, \dots, x_{10}$ such that $\sum_{i=1}^{10} (x_i - \alpha) = 2$ and $\sum_{i=1}^{10} (x_i - \beta)^2 = 40$, where α, β are positive integers. Let the mean and the variance of the observations be $\frac{6}{5}$ and $\frac{84}{25}$ respectively. The $\frac{\beta}{\alpha}$ is equal

to :

- (1) 2
- (2) $\frac{3}{2}$
- (3) $\frac{5}{2}$
- (4) 1

Q3 - 2024 (27 Jan Shift 1)

Let $a_1, a_2, \dots, a_{10}$ be 10 observations such that $\sum_{k=1}^{10} a_k = 50$ and $\sum_{k < j} a_k \cdot a_j = 1100$. Then the standard deviation of $a_1, a_2, \dots, a_{10}$ is equal to :

- (1) 5
- (2) $\sqrt{5}$
- (3) 10
- (4) $\sqrt{115}$

Q4 - 2024 (27 Jan Shift 2)

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The mean and standard deviation of 15 observations were found to be 12 and 3 respectively. On rechecking it was found that an observation was read as 10 in place of 12. If μ and σ^2 denote the mean and variance of the correct observations respectively, then $15(\mu + \mu^2 + \sigma^2)$ is equal to

Q5 - 2024 (29 Jan Shift 1)

If the mean and variance of the data 65, 68, 58, 44, 48, 45, 60, α , β , 60 where $\alpha > \beta$ are 56 and 66.2 respectively, then $\alpha^2 + \beta^2$ is equal to

Q6 - 2024 (29 Jan Shift 2)

If the mean and variance of five observations are $\frac{24}{5}$ and $\frac{194}{25}$ respectively and the mean of first four observations is $\frac{7}{2}$, then the variance of the first four observations is equal to

- (1) $\frac{4}{5}$
- (2) $\frac{77}{12}$
- (3) $\frac{5}{4}$
- (4) $\frac{105}{4}$

Q7 - 2024 (30 Jan Shift 1)

Let M denote the median of the following frequency distribution.

Class	0 – 4	4 – 8	8 – 12	12 – 16	16 – 20
Frequency	3	9	10	8	6

Then $20M$ is equal to :

- (1) 416
- (2) 104
- (3) 52
- (4) 208

Q8 - 2024 (30 Jan Shift 2)

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The variance σ^2 of the data

x_i	0	1	5	6	10	12	17
f_i	3	2	3	2	6	3	3

Is _____.

Q9 - 2024 (31 Jan Shift 2)

Let the mean and the variance of 6 observation $a, b, 68, 44, 48, 60$ be 55 and 194, respectively if $a > b$, then

$a + 3b$ is

(1) 200

(2) 190

(3) 180

(4) 210

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Questions with Answer Keys

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Answer Key

Answer Key

Q1 (3) mathongo // math Q2 (1) // math Q3 (2) math Q4 (2521) // math

Q5 (6344) **Q6** (3) **Q7** (4) **Q8** (29)

Q9 (3) mathongo // mathongo // mathongo // mathongo // mathongo // mathongo

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Solutions

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Q1

$$\text{Median} = 170 \Rightarrow 125, a, b, 170, 190, 210, 230$$

Mean deviation about

Median =

$$\frac{0 + 45 + 60 + 20 + 40 + 170 - a + 170 - b}{7} = \frac{205}{7}$$

$$\Rightarrow a + b = 300$$

$$\text{Mean} = \frac{170 + 125 + 230 + 190 + 210 + a + b}{7} = 175$$

Mean deviation

About mean =

$$\frac{50 + 175 - a + 175 - b + 5 + 15 + 35 + 55}{7} = 30$$

Q2

$$x_1, x_2, \dots, x_{10}$$

$$\sum_{i=1}^{10} (x_i - \alpha) = 2 \Rightarrow \sum_{i=1}^{10} x_i - 10\alpha = 2$$

$$\text{Mean } \mu = \frac{6}{5} = \frac{\sum x_i}{10}$$

$$\therefore \sum x_i = 12$$

$$10\alpha + 2 = 12 \therefore \alpha = 1$$

$$\text{Now } \sum_{i=1}^{10} (x_i - \beta)^2 = 40 \quad \text{Let } y_i = x_i - \beta$$

$$\therefore \sigma_y^2 = \frac{1}{10} \sum y_i^2 - (\bar{y})^2$$

$$\sigma_x^2 = \frac{1}{10} \sum (x_i - \beta)^2 - \left(\frac{\sum_{i=1}^{10} (x_i - \beta)}{10} \right)^2$$

$$\frac{84}{25} = 4 - \left(\frac{12 - 10\beta}{10} \right)^2$$

$$\therefore \left(\frac{6 - 5\beta}{5} \right)^2 = 4 - \frac{84}{25} = \frac{16}{25}$$

$$6 - 5\beta = \pm 4 \Rightarrow \beta = \frac{2}{5} \text{ (not possible) or } \beta = 2$$

$$\text{Hence } \frac{\beta}{\alpha} = 2$$

Q3

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Solutions

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$$\sum_{k=1}^{10} a_k = 50$$

$$a_1 + a_2 + \dots + a_{10} = 50 \dots (i)$$

$$\sum_{\forall k < j} a_k a_j = 1100 \dots (ii)$$

$$\text{If } a_1 + a_2 + \dots + a_{10} = 50.$$

$$(a_1 + a_2 + \dots + a_{10})^2 = 2500$$

$$\Rightarrow \sum_{i=1}^{10} a_i^2 + 2 \sum_{k < j} a_k a_j = 2500$$

$$\Rightarrow \sum_{i=1}^{10} a_i^2 = 2500 - 2(1100)$$

$$\sum_{i=1}^{10} a_i^2 = 300, \text{ Standard deviation ' } \sigma \text{ '}$$

$$\begin{aligned} &= \sqrt{\frac{\sum a_i^2}{10} - \left(\frac{\sum a_i}{10}\right)^2} = \sqrt{\frac{300}{10} - \left(\frac{50}{10}\right)^2} \\ &= \sqrt{30 - 25} = \sqrt{5} \end{aligned}$$

Q4

Let the incorrect mean be μ' and standard deviation be σ'

We have

$$\mu' = \frac{\sum x_i}{15} = 12 \Rightarrow \sum x_i = 180$$

As per given information correct $\sum x_i = 180 - 10 + 12$

$$\Rightarrow \mu(\text{correct mean}) = \frac{182}{15}$$

Also

$$\sigma' = \sqrt{\frac{\sum x_i^2}{15} - 144} = 3 \Rightarrow \sum x_i^2 = 2295$$

$$\text{Correct } \sum x_i^2 = 2295 - 100 + 144 = 2339$$

$$\sigma^2(\text{correct variance}) = \frac{2339}{15} - \frac{182 \times 182}{15 \times 15}$$

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Solutions

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Required value

$$\begin{aligned}
 &= 15 (\mu + \mu^2 + \sigma^2) \\
 &= 15 \left(\frac{182}{15} + \frac{182 \times 182}{15 \times 15} + \frac{2339}{15} - \frac{182 \times 182}{15 \times 15} \right) \\
 &= 15 \left(\frac{182}{15} + \frac{2339}{15} \right) \\
 &= 2521
 \end{aligned}$$

Q5

$$\bar{x} = 56$$

$$\sigma^2 = 66.2$$

$$\Rightarrow \frac{\alpha^2 + \beta^2 + 25678}{10} - (56)^2 = 66.2$$

$$\therefore \alpha^2 + \beta^2 = 6344$$

Q6

$$\bar{X} = \frac{24}{5}; \sigma^2 = \frac{194}{25}$$

Let first four observation be x_1, x_2, x_3, x_4

$$\text{Here, } \frac{x_1 + x_2 + x_3 + x_4 + x_5}{5} = \frac{24}{5} \dots$$

$$\text{Also, } \frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{7}{2}$$

$$\Rightarrow x_1 + x_2 + x_3 + x_4 = 14$$

Now from eqn -1

$$x_5 = 10$$

$$\text{Now, } \sigma^2 = \frac{194}{25}$$

$$\frac{x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2}{5} - \frac{576}{25} = \frac{194}{25}$$

$$\Rightarrow x_1^2 + x_2^2 + x_3^2 + x_4^2 = 54$$

Now, variance of first 4 observations

$$\text{Var} = \frac{\sum_{i=1}^4 x_i^2}{4} - \left(\frac{\sum_{i=1}^4 x_i}{4} \right)^2$$

$$= \frac{54}{4} - \frac{49}{4} = \frac{5}{4}$$

Q7

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Solutions

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Class	Frequency	Cumulative frequency
0 – 4	3	3
4 – 8	9	12
8 – 12	10	22
12 – 16	8	30
16 – 20	6	36

$$M = 1 + \left(\frac{\frac{N}{2} - C}{f} \right) h$$

$$M = 8 + \frac{18 - 12}{10} \times 4$$

$$M = 10.4$$

$$20M = 208$$

Q8

x_i	f_i	$f_i x_i$	$f_i x_i^2$
0	3	0	0
1	2	2	2
5	3	15	75
6	2	12	72
10	6	60	600
12	3	36	432
17	3	51	867
	$\Sigma f_i = 22$		$\Sigma f_i x_i^2 = 2048$

$$\therefore \Sigma f_i x_i = 176$$

$$\text{So } \bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{176}{22} = 8$$

$$\text{for } \sigma^2 = \frac{1}{N} \Sigma f_i x_i^2 - (\bar{x})^2$$

$$= \frac{1}{22} \times 2048 - (8)^2$$

$$= 93.090964$$

$$= 29.0909$$

Q9

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Solutions

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 $a, b, 68, 44, 48, 60$

Mean = 55

Variance = 194 $a > b$

$$\frac{a + b + 68 + 44 + 48 + 60}{6} = 55$$

$$\Rightarrow 220 + a + b = 330$$

$$\therefore a + b = 110 \dots (1)$$

Also,

$$\sum \frac{(x_i - \bar{x})^2}{n} = 194$$

$$\Rightarrow (a - 55)^2 + (b - 55)^2 + (68 - 55)^2 + (44 - 55)^2$$

$$+ (48 - 55)^2 + (60 - 55)^2 = 194 \times 6$$

$$\Rightarrow (a - 55)^2 + (b - 55)^2 + 169 + 121 + 49 + 25 = 1164$$

$$\Rightarrow (a - 55)^2 + (b - 55)^2 = 1164 - 364 = 800$$

$$a^2 + 3025 - 110a + b^2 + 3025 - 110b = 800$$

$$\Rightarrow a^2 + b^2 = 800 - 6050 + 12100$$

$$a^2 + b^2 = 6850 \dots (2)$$

Solve (1) & (2);

$$a = 75, b = 35$$

$$\therefore a + 3b = 75 + 3(35) = 75 + 105 = 180$$

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